



Repo and the Retirement of **LIBOR**

*the plumbing beneath the price of money,
and the benchmark that got caught rigging it*

RATTY2TAILS

A short history of short-term rates

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Repo and the Retirement of **LIBOR**

On the market that prices money overnight, and the benchmark that priced it by asking politely

TL;DR

- The **repo market** is where banks and other institutions borrow and lend cash overnight (or for a few days) against government bonds and similar collateral. It is the least glamorous, most essential market in the financial system — the overnight plumbing that keeps everyone's cash and collateral in the right place each morning.
- In **September 2019**, that plumbing seized up: overnight repo rates briefly spiked to nearly 10%, more than four times the Fed's target, forcing the Federal Reserve back into daily market operations for the first time since the 2008 crisis.
- **LIBOR** (the London Interbank Offered Rate) was, for decades, the world's benchmark interest rate — underpinning hundreds of trillions of dollars in loans, mortgages, and derivatives. It was not measured from real transactions; it was *submitted* daily by a panel of banks estimating what they thought they'd pay to borrow.
- That estimate-based design made it gameable. The **2012 LIBOR scandal** revealed traders at major banks colluding to nudge submissions in whichever direction suited their trading books — one of the largest rate-rigging conspiracies in financial history.
- Regulators spent the following decade retiring it. **USD LIBOR panel submissions ended on 30 June 2023**, replaced by **SOFR** (the Secured Overnight Financing Rate) — a benchmark built directly from actual, observable repo transactions rather than bank estimates.
- The through-line: the new benchmark is only as trustworthy as the market beneath it — which is exactly why understanding repo matters before understanding what replaced LIBOR.

§1 Introduction and Scope

There are two ways to tell the story of how the world prices short-term money. One is the story of a scandal — traders in Canary Wharf and Manhattan trading instant-messages about nudging a number a few basis points in their favour, regulators discovering it, and a benchmark trusted since 1986 being walked out of the building in disgrace. The other is the story of plumbing — reserves, collateral, haircuts, and the unglamorous overnight market where institutions actually borrow and lend cash against government

bonds, a market almost nobody outside finance has heard of and almost nobody inside finance can do without.

This essay tells both, because they are, in the end, the same story. LIBOR was retired not merely because it was rigged, but because it was rigged in a way that was structurally inevitable — it asked banks what they *thought* borrowing would cost rather than measuring what borrowing actually *did* cost. Its replacement, SOFR, was built to close that gap by anchoring the world's most important interest rate to the repo market: real trades, real collateral, real cash, tens of billions of dollars deep every single night. Understanding why that substitution mattered means understanding repo first.

The essay proceeds in three movements. Sections 2 through 4 explain what the repo market is, why it sits underneath almost everything else in the financial system, and what happened when it broke in September 2019. Sections 5 through 7 turn to LIBOR: what it was for, how it was gamed, and the decade-long process of retiring it. Section 8 explains SOFR and how it is built from repo data. Section 9 closes with the reading this series keeps returning to — what a benchmark's design says about the monetary system it serves, and what it means that the system's most important price still depends, in the end, on a central bank standing by with a facility to make sure the plumbing never floods again.

§2 What the Repo Market Actually Is

A repurchase agreement — a "repo" — is, underneath the jargon, a fully collateralised overnight loan dressed up as two trades. Party A sells a government bond to Party B today, for cash, and simultaneously agrees to buy that same bond back tomorrow at a slightly higher price. The difference between the sale price and the repurchase price, annualised, is the interest rate on the loan — the **repo rate**. Structuring it as a sale-and-repurchase rather than a straightforward secured loan matters legally: if Party A defaults overnight, Party B already owns the bond outright and can sell it immediately, without wading through bankruptcy proceedings to seize collateral.

From Party A's side, this is a way to borrow cash cheaply against bonds it already owns, without selling them outright — useful for a dealer who needs overnight cash but wants to keep its bond position for tomorrow. From Party B's side — often a money-market fund, a corporate treasurer, or another bank with idle cash — this is a way to earn interest overnight with virtually no credit risk, since the loan is backed by a government bond and typically over-collateralised through a **haircut** (Party B might lend only \$98 in cash against \$100 of bonds, so a small overnight price move doesn't leave it under-secured).

~\$4tn+

ESTIMATED DAILY VOLUME, U.S. REPO
MARKET

Tri-party, bilateral, and centrally-cleared repo
combined, by mid-2020s Fed/OFR estimates.

O/N

TYPICAL MATURITY: OVERNIGHT

Most repo is rolled every single business day — a
market that resets itself daily.

Two broad flavours dominate. In **general collateral (GC) repo**, the borrower can post more or less any acceptable Treasury security and the rate reflects the general price of cash overnight. In a **specials** repo, the cash lender wants one *particular* bond — often because it needs to cover a short position in that exact security — and will lend cash at a below-market rate just to get its hands on it. GC repo rates track the broad cost of overnight funding; specials rates reveal which individual bonds are, at that moment, in short supply. Repo also comes in two settlement styles: **bilateral**, where two parties settle directly, and **tri-party**, where a clearing bank (in the U.S., overwhelmingly the Bank of New York Mellon) sits in the middle, holding collateral and managing settlement for both sides — the arrangement that lets the market handle its enormous daily volume without every desk having to independently verify every bond it receives.

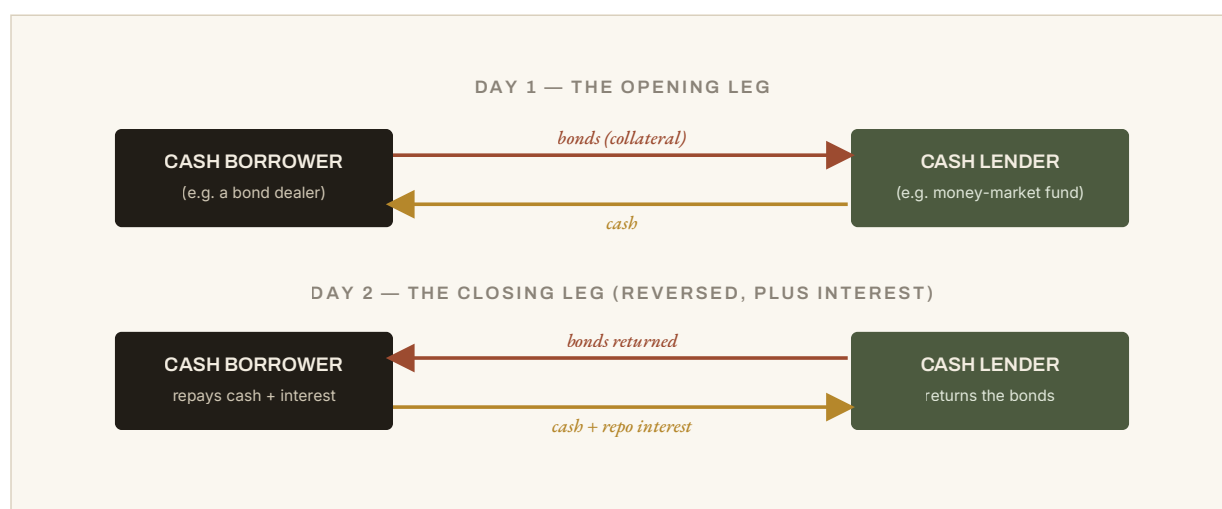


FIG. 1 — A single overnight repo, unwound. Bonds move one way, cash the other, then both legs reverse the next morning with a small interest payment folded into the repurchase price.

§3 Repo Is Not a Sideshow — It Is the Basement

It is tempting, from outside finance, to file repo alongside other obscure market plumbing and move on. That would be a mistake. Repo is the mechanism by which the largest, most liquid asset in the world — government debt — is turned back into the second-largest asset in the world, cash, on a nightly basis. Primary dealers finance their entire Treasury bond

inventories through repo. Hedge funds use repo to lever up positions many times over, borrowing cash against bonds, buying more bonds, repo-ing those too. Money-market funds — the vehicles that hold the "safe" cash of corporations, pension funds, and retail savers — park a large share of their assets in overnight repo precisely because it is collateralised, short-dated, and (in normal times) essentially risk-free.

Central banks depend on it too, in both directions. The Federal Reserve's own toolkit for steering short-term interest rates works substantially through repo and its mirror image, **reverse repo** (where the Fed itself borrows cash from money funds against Treasuries, draining excess liquidity). When the Fed wants to inject reserves into the banking system quickly, it can do so via repo operations; when it wants to mop up excess cash, it runs reverse repo. The overnight reverse repo facility (ON RRP) became, in the post-2020 era of enormous excess reserves, one of the Fed's single largest and most closely watched policy tools — often absorbing well over a trillion dollars a night at its peak.

WHY "GENERAL COLLATERAL" MATTERS

In GC repo, the specific bond posted barely matters — any acceptable Treasury will do, and the rate simply reflects the going price of overnight cash. This is precisely why GC repo rates make good raw material for a benchmark: they are a clean read on the market-wide cost of secured overnight borrowing, uncontaminated by any single bond's idiosyncratic scarcity.

This is also why repo rates are watched so closely as a stress indicator. Because repo sits at the intersection of bank balance sheets, dealer inventories, and money-fund cash, a sudden spike in overnight repo rates is rarely about repo itself — it is a symptom of something upstream: too many bonds needing financing at once, too few institutions willing or able to supply the cash, or reserves in the banking system simply running short. Which is exactly what happened in the autumn of 2019.

§4 September 2019: When the Basement Flooded

On the morning of Monday 16 September 2019, the overnight GC repo rate — normally trading within a whisker of the Fed's target range — spiked toward **10%**, more than quadruple where it had closed the previous week. The Fed's own reference rate for the market, the effective federal funds rate, was dragged up alongside it, briefly trading above the top of the Fed's target band — an uncomfortable look for a central bank whose entire job, in that moment, was supposed to be keeping that rate pinned in place.

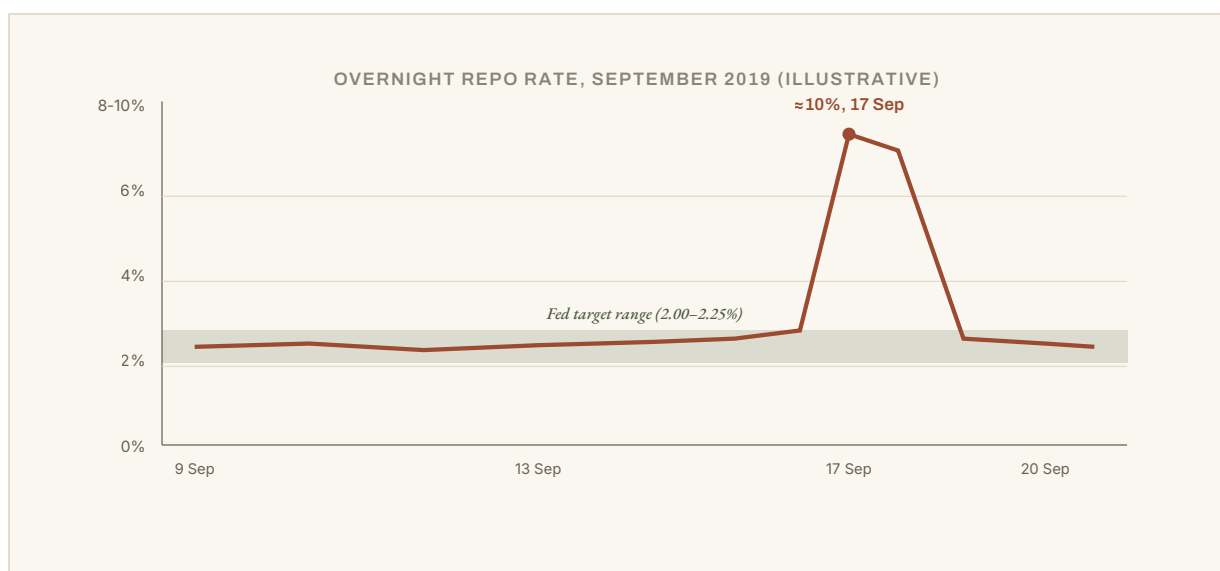


FIG. 2 — The overnight repo rate briefly tore free of the Fed's target band before the New York Fed's repo operations pulled it back down. Illustrative reconstruction of the shape of the spike, not tick-level data.

The immediate triggers were mundane and, with hindsight, entirely predictable: a corporate tax deadline was pulling cash out of the banking system just as a large Treasury bond settlement was pulling in the opposite direction, both landing on the same day. But the reason a routine calendar coincidence turned into a market seizure was structural. Since the 2008 crisis, post-crisis regulation (notably Basel III liquidity rules) had made large banks reluctant to deploy their balance sheets into repo lending even when they technically held ample reserves, because doing so consumed regulatory capital and liquidity-coverage headroom they preferred to keep in reserve. At the same time, years of quantitative tightening had been steadily shrinking the pool of reserves in the banking system. Reserves that had once been abundant had, by September 2019, become merely adequate — and adequate reserves, it turned out, were not enough to absorb a one-off cash squeeze.

The New York Fed responded by injecting tens of billions of dollars a day into the repo market through repeated operations, and within about a week had restored order. But the episode did lasting damage to a comfortable assumption: that the Fed could manage short-term rates passively, through the interest it paid on reserves, without needing to actively intervene in repo markets. Instead, the Fed went on to establish a **Standing Repo Facility (SRF)** in 2021 — a permanent, always-on backstop letting eligible institutions borrow cash against Treasury and agency collateral whenever the market rate threatens to run away, precisely so that September 2019 does not need to repeat itself. It is worth sitting with what that backstop represents: the modern financial system's most important overnight market cannot, on the regulators' own admission, be trusted to clear itself in a crunch without a standing central-bank facility underneath it.

§5 What LIBOR Was, and Why It Existed

Wind the clock back to 1986. Banks needed a common reference rate for the interbank lending market — a single number that a floating-rate loan, a swap, or a mortgage could be priced "off," so that a borrower and a lender could agree a rate of "LIBOR plus 150 basis points" without either side needing to negotiate the base rate itself every time. The British Bankers' Association built exactly that: each morning, a panel of major banks in London would submit the rate at which *they believed* they could borrow unsecured funds from other banks, for a range of maturities and currencies. The submissions were trimmed — the highest and lowest thrown out — and averaged into a single published number: the London Interbank Offered Rate.

It worked, in the sense that it spread everywhere. By the 2000s, LIBOR (and its currency-specific siblings, including sterling and euro variants) underpinned an estimated **\$300–400 trillion** in financial contracts worldwide — adjustable-rate mortgages, syndicated corporate loans, student loans, and above all interest-rate derivatives, whose notional value dwarfed the "real economy" contracts many times over. A benchmark of that scale needed to be dependable in the dullest possible way. Its actual design was anything but.

THE DESIGN FLAW, STATED PLAINLY

LIBOR was not measured. It was asked. Panel banks reported what they believed they could borrow at — not what they had actually paid, transaction by transaction — which meant the number depended, every single day, on the honesty of a small group of self-interested submitters with no external check on their answers.

Two structural weaknesses sat underneath that design. First, especially after the 2008 crisis, the unsecured interbank lending market that LIBOR was supposed to describe had thinned dramatically — banks simply weren't lending to each other unsecured in the volumes they once had, so submitters were increasingly estimating a rate for a market that barely existed. Second, and more corrosively, a submission was a private number reported by a single desk at a single bank, checked against nothing but the trimmed-mean process itself. There was no requirement to show your work.

§6 The Rigging: How a Benchmark Built on Guesswork Was Gamed

Where there is a number that moves trillions of dollars in derivatives payouts and can be nudged by a phone call to the desk next door, someone will eventually make that phone call. From the mid-2000s through the financial crisis, traders at multiple major banks — including Barclays, UBS, Royal Bank of Scotland, Rabobank, and Deutsche Bank, among others — were found to have coordinated with their own banks' LIBOR submitters, asking

them to submit rates a few basis points higher or lower than their honest estimate, purely because it would move the value of a derivatives book the trader was holding in the bank's or a client's favour.

A second, separate distortion emerged during the crisis itself: some banks, worried that an honest (higher) LIBOR submission would be read by the market as a signal of their own funding stress, submitted artificially low rates to appear healthier than they were — a form of rigging aimed not at derivatives profit but at reputation management during a period when confidence in bank solvency was already fragile.

YEAR	DEVELOPMENT
2008	Wall Street Journal analysis and internal Barclays communications raise early public questions about whether LIBOR submissions reflect genuine borrowing costs.
2012	Regulatory investigations across the UK and US become public; Barclays settles first, followed by UBS, RBS, and others, admitting to submission manipulation for trading gain and reputational reasons.
2013–2015	Criminal prosecutions of individual traders and brokers follow in the UK and US; the British Bankers' Association is stripped of oversight, with administration handed to ICE Benchmark Administration.
2017	The UK's Financial Conduct Authority announces panel banks will no longer be compelled to submit LIBOR rates after end-2021, effectively setting a retirement date.
2021	Publication of most LIBOR settings ceases (or becomes "synthetic" and non-representative) at end-December; USD LIBOR is granted an 18-month extension for legacy contracts.
2023	USD LIBOR panel-based publication ends permanently on 30 June, closing the book on the benchmark entirely.

Regulators on both sides of the Atlantic ultimately levied billions of dollars in fines against the implicated banks, and a handful of traders were convicted of fraud-related offences. But punishing the individuals who gamed a flawed system does not fix the flaw. Regulators concluded, correctly, that no amount of enforcement could make a benchmark built on unverifiable private estimates fully trustworthy — the honest solution was to stop asking banks what they thought, and start measuring what actually happened in a market deep and transparent enough to not need asking.

§7 The Long Retirement: From Scandal to Cessation

The gap between "this benchmark is untrustworthy" (2012) and "this benchmark no longer exists" (2023) spans over a decade, and that gap is itself instructive. Trillions of dollars in

outstanding loans, bonds, and derivatives referenced LIBOR at any given moment; unwinding that dependency without triggering chaotic repricing, litigation, or contract disputes required years of coordinated work by regulators, industry working groups, and legislators.

In the US, the Alternative Reference Rates Committee (ARRC), convened by the Federal Reserve, spent years building market infrastructure and fallback language for the new benchmark before the transition date arrived. Where legacy contracts had no workable fallback language of their own — old mortgages and bonds drafted long before anyone anticipated LIBOR's death — the US Congress ultimately passed the **LIBOR Act** in 2022, imposing a statutory replacement rate on contracts that would otherwise have been left referencing a benchmark that no longer existed. The UK ran a parallel process through the FCA, with sterling markets migrating to SONIA (the Sterling Overnight Index Average) well ahead of the US timeline.

§8 Enter SOFR: A Rate Made of Repo

The rate chosen to replace USD LIBOR was not an improved version of the same idea. It was a different kind of number entirely — one built from precisely the market described in Sections 2 through 4 of this essay. The **Secured Overnight Financing Rate (SOFR)**, published daily by the Federal Reserve Bank of New York, is calculated from actual transaction data across three segments of the Treasury repo market: tri-party repo, General Collateral Finance (GCF) repo, and bilaterally cleared repo. Rather than asking a panel of banks what they think borrowing costs, the New York Fed simply looks at what tens of thousands of real overnight repo transactions — often exceeding a trillion dollars of volume on an average day — actually traded at, and calculates a volume-weighted median.

Estimate

LIBOR, BY DESIGN

A panel bank's opinion of its own unsecured borrowing cost, self-reported and unverified against transaction records.

Transaction

SOFR, BY DESIGN

A volume-weighted median of actual overnight Treasury repo trades — secured, observable, and enormous in scale.

This design closes the manipulation channel that sank LIBOR almost by construction: there is no submission to nudge, because there is no submission. A trader wishing to move SOFR would need to move the price of a market transacting well over a trillion dollars a day — several orders of magnitude harder than a phone call between two desks. The trade-off is that SOFR measures something subtly different from what LIBOR measured. LIBOR was an *unsecured* interbank rate, carrying a small credit-risk premium reflecting the possibility

that the borrowing bank might default; SOFR is a *secured* rate, backed by Treasury collateral, and so sits structurally lower and behaves differently under stress — precisely the September 2019 kind of stress, when secured repo rates can spike sharply even as unsecured bank credit risk barely moves. Markets have spent the years since the transition building new tools — including forward-looking term SOFR rates — to recreate some of the practical conveniences LIBOR offered without reintroducing its central flaw.

§9 The Ratty2Tails Reading: Benchmarks, Backstops, and Bearer Assets

The LIBOR story and the repo story are, together, a case study in something this series returns to repeatedly: the difference between a price that is *declared* and a price that is *discovered*. LIBOR failed not because bankers are uniquely dishonest — though some plainly were — but because its design invited the failure. Any price-setting mechanism that depends on a small number of interested parties self-reporting, unverified, into a number that moves trillions of dollars, will eventually be gamed. That is not a moral claim about bankers; it is closer to a law of incentives.

"A rate that has to be asked for is a rate someone can be asked to change."

— ON THE STRUCTURAL DIFFERENCE BETWEEN LIBOR AND SOFR

SOFR's answer — anchor the number to a real, deep, transparently observable market — is a genuine improvement, and it is not a coincidence that regulators reached for something closer to a market-discovered price after the scandal made the alternative untenable. But it is worth being precise about how far that improvement goes. SOFR is harder to manipulate than LIBOR; it is not independent of the state. It is administered by the Federal Reserve Bank of New York, built from a market whose stability the Fed itself now explicitly backstops through the Standing Repo Facility, and calculated within a monetary system where the supply of the underlying reserves is itself a policy choice, made by committee, revisable at will. Repo rates are more honest than LIBOR submissions. They are not, and were never designed to be, independent of central bank discretion — September 2019 demonstrated that the market cannot even be trusted to clear itself without one.

This is precisely the axis on which this series has repeatedly measured Bitcoin against the instruments and institutions of the existing monetary order. LIBOR was a price set by a small, self-interested committee with no external verification — exactly the failure mode a fixed-supply, permissionlessly-verifiable bearer asset is structurally incapable of replicating, because there is no panel, no submission, and no administrator to lean on. SOFR is a genuine step toward market-discovered pricing, built from real transactions rather than self-reported guesses — and it is precisely because that step was worth taking that the fur-

ther step is worth naming: a market whose deepest, most important interest rate still requires a standing central-bank facility to guarantee its own liquidity is not a market pricing money independently of the state. It is a market pricing money with the state's hand permanently resting near the tap, ready to open it the moment the basement floods again.

— Summary

The repo market is the overnight engine room of the financial system: an enormous, largely invisible market where cash and government bonds are exchanged nightly, collateralising trillions of dollars of borrowing at rates most people never see. In September 2019, a routine cash-flow coincidence exposed how thin the margin for error in that market had become, forcing the Federal Reserve back into daily intervention and eventually into a standing backstop facility.

LIBOR, for decades the world's reference interest rate, was undone by a much older problem: it measured belief, not transactions, and belief proved gameable. The 2012 rigging scandal made that flaw impossible to ignore, and by 30 June 2023 the benchmark was gone entirely, replaced in US dollars by SOFR — a rate built directly from real repo transactions rather than bank submissions. It is a genuinely more honest number. But it is discovered inside a market whose own stability the central bank now explicitly guarantees, which is a useful reminder that "market-based" and "independent of the state" are not the same claim, and rarely have been, in the architecture of fiat money.

— References

1. Federal Reserve Bank of New York — Secured Overnight Financing Rate (SOFR) data and methodology.
2. Federal Reserve Bank of New York — "Reflections on the 2019 repo market disruption," staff reports and remarks.
3. Board of Governors of the Federal Reserve System — establishment and operation of the Standing Repo Facility (2021).
4. UK Financial Conduct Authority — statements on LIBOR panel cessation and the transition timeline (2017–2023).
5. Alternative Reference Rates Committee (ARRC) — publications on the LIBOR-to-SOFR transition.
6. US Department of Justice and Commodity Futures Trading Commission — settlement and enforcement documents relating to LIBOR manipulation, Barclays, UBS, RBS, and others (2012–2015).

7. Adjustable Interest Rate (LIBOR) Act, Pub. L. 117-328 (2022), and implementing regulations.
8. Bank for International Settlements — repo market and reference rate reform publications.

The Ratty2Tails Papers — Monetary History Series